

SIH26069 — National Weather Big Data Analytics Platform

Comprehensive OSINT & GitHub Research Report

1. Executive Summary

An existing public implementation of SIH26069 does not exist. No repository, paper, or demo explicitly implements 'SIH26069 — National Weather Big Data Analytics Platform' as a complete end-to-end system matching the official problem statement.

Approximately 25–30 public repositories implement substantial overlapping components: real-time weather streaming (Kafka/Spark/Airflow), weather ETL pipelines, weather dashboards (React/Flask/Next.js), extreme weather classification (ML), fake news/misinformation detection (NLP), GIS weather maps (Leaflet/Mapbox/PostGIS), IMD API wrappers and Indian weather datasets.

No single repo covers >70% of SIH26069 requirements. Estimated 40–60% of functional blocks exist separately across multiple repos.

2. Best Matching Projects (Top 10)

Rank	Project	Similarity	Completeness	Main Tech
1	kafka-spark-cassandra	75%	Full pipeline	Python, Kafka, PySpark, Cassandra
2	openweather-data-pipeline	70%	Full ETL	Python, Airflow, PySpark, Azure SQL
3	DripTect (Flood Guard)	68%	Full-stack	ESP32, Flask, Next.js, OpenWeather API, RF ML
4	Weather-DAG-Airflow	65%	Airflow ETL	Python, Airflow, AWS, Postgres
5	Weather-Data-Integration	63%	ETL with Spark	Python, Spark, Airflow, Postgres
6	Extreme-Weather-Events-Classification	60%	ML classification	Python, scikit-learn, XGBoost
7	SkySense Weather Dashboard	58%	React dashboard	React, Azure Functions, Redis
8	Weather-Dashboard-Station	55%	React + Leaflet	React, Vite, Leaflet, Chart.js
9	Fake_news_detector (Climate)	50%	ML classifier	Python, scikit-learn, Flask
10	GDSC-Group-Project (SIH Air/Water)	48%	SIH environmental	Python, Flask, React

3. Recommended Architecture

DATA SOURCES → DATA INGESTION (Python producers, REST APIs) → STREAM PROCESSING (Apache Kafka) → AI/ML (Fake Detection, Event Classification) + Deduplication (Semantic Similarity, FAISS) → DATA STORAGE (PostgreSQL + PostGIS, Delta Lake, Elasticsearch) → ANALYTICS ENGINE (PySpark, Pandas) → GIS / DASHBOARD (Leaflet/Mapbox, Next.js, Grafana) → ADMIN PANEL (Verification Workflow, User Management)

4. Technology Stack

Layer	Technology	Evidence
Frontend	Next.js + Leaflet + Recharts	DripTect dashboard

Backend	FastAPI + Celery	Common in weather ETL repos
Database	PostgreSQL + PostGIS	Standard for GIS weather apps
Streaming	Apache Kafka + Schema Registry	kafka-spark-cassandra
Big Data	PySpark Structured Streaming	kafka-spark-cassandra, weather-data-integration
ML/AI	Hugging Face Transformers + scikit-learn	Extreme-Weather-Classification, Fake_news_detector
GIS	PostGIS + Leaflet	Weather-Dashboard-Station
Storage	Delta Lake (raw) + PostGIS (query)	openweather-data-pipeline
Deployment	Docker + Kubernetes	kafka-spark-cassandra Docker setup

5. Key Repository Links

kafka-spark-cassandra: <https://github.com/debanjansaha-git/kafka-spark-cassandra>

openweather-data-pipeline: <https://github.com/erreduarte/openweather-data-pipeline>

DripTect (Flood Guard): https://github.com/anshley/DripTect_Statuscode-2.0

Weather-DAG-Airflow: <https://github.com/VincentRevel/Weather-DAG-Airflow>

weather-data-integration: <https://github.com/bitsbard/weather-data-integration>

Extreme-Weather-Events-Classification: <https://github.com/chiragoel/Extreme-Weather-Events-Classification>

SkySense Dashboard: <https://github.com/miguelbtcode/weather-dashboard>

Weather-Dashboard-Station: <https://github.com/khushi-purwar/Weather-Dashboard-Station>

Fake_news_detector (Climate): https://github.com/AMustafa4983/Fake_news_detector

GDSC-Group-Project (SIH Air/Water): <https://github.com/Gauranshgoel123/GDSC-Group-Project>

6. Final Verdict

NO exact match exists, but ~25–30 public repositories implement overlapping components. You can reuse 40–60% of the architecture by combining: - kafka-spark-cassandra (Kafka + PySpark streaming) - openweather-data-pipeline (Airflow ETL) - DripTect (disaster monitoring + ML + Next.js dashboard) - Extreme-Weather-Events-Classification (event classification) - Fake_news_detector (fake detection) Build the missing 40% (social media mining, citizen reports, verification workflow, GIS heatmaps, multilingual support) to create a unique, India-focused SIH26069 solution.